PREETHAM PRATHIPATI

617-206-0361 | preeth@bu.edu | linkedin.com/in/preethamprathipati | github.com/artisticdrake | preethamprathipati.com

SUMMARY

ML Engineer graduating with an MS in Computer Science, Boston University (May 2026). Won $4,000 at the Bio-IT World Hackathon 2026 for an interpretable cancer outcome prediction system built with PyTorch and AWS Bedrock. Fine-tuned BERT to 98% NER accuracy and deployed an LLM-powered product with RAG and deterministic pipelines, serving 30+ users. Seeking to leverage strong ML research and production experience to deliver high-impact AI solutions.

AWARDS & RECOGNITION

- **Bio-IT World Hackathon 2026 - Prize Winner, Fair Data Track:**$4,000 Award · Team of 4 · 6 Teams Competed May 2026 - Built interpretable Visible Neural Network predicting cancer patient outcomes from tumor genotype on NIH CFDE datasets; judges awarded excellence in model interpretability
- **National ML Ranking - 2nd of 364:**Foundations of Modern Machine Learning, IIIT Hyderabad 2023 - Top 0.6% nationally across 364 participants in competitive ML certification program

PROJECTS

NeST-VNN: Interpretable Cancer Outcome Prediction — **May 2026**

- Architecture: Hierarchical Visible Neural Network mirroring biological ontology (NeST hierarchy) - each node is a PyTorch protein assembly module, making predictions interpretable at pathway level via RLIPP explainability scores rather than black-box gene-level outputs
- End-to-end pipeline: cBioPortal genomic ingestion (mutations, CNV, fusions) → feature transformation → PyTorch training (AdamW, early stopping, gradient masking for ontology sparsity) → MLflow experiment tracking across 20 Optuna trials → RLIPP pathway annotation
- LLM integration: Built custom RAG pipeline with SQLite metrics store feeding AWS Bedrock to generate plain-English biological pathway interpretations for clinical users; built NeST-STRING Explorer (Flask + Cytoscape.js) with per-patient mutation, copy-number, and fusion visualization against STRING PPI networks
- Result: Won $4,000 prize; judges cited excellence in interpretability; work builds on Nature Cancer (2024) and Cancer Discovery (2024) methodology

UniGuide - Intelligent Campus NLP Assistant — **2024 - 2025**

- Model: Fine-tuned BERT-based NER on 3,500+ custom-labeled campus queries across 6 intent classes - improved intent accuracy from 71% (baseline spaCy) to 98% (+27pp); 96% semantic routing precision - graduate Software Engineering capstone
- System: 9-stage NLP pipeline with Retrieval-Augmented Generation (RAG), FAISS vector search, RapidFuzz session memory, and dynamic LLM routing (Ollama/Qwen local ↔ OpenAI GPT); deployed to 30+ real BU campus users
- Infrastructure: FastAPI microservices backend, PostgreSQL vector storage, Docker + Jenkins CI/CD; Redis caching reduced query latency by 35%

Zenith - AI-Powered Application Tracker — **2024**

- Product: Full-stack Webapp (React/Vite + Node.js/Express + Supabase + Google OAuth) actively used to manage 100+ job applications; built resume builder tab with Jake's Resume live preview, LaTeX export, and one-page auto-fit
- AI pipeline: Deterministic resume–JD matching engine (200-skill canonical dictionary, regex section parsing, implied skill inference) replacing unreliable LLM scoring; GPT-4o autofill extracting structured job data from raw URLs; "Zenith" AI summary panel for job-fit analysis; SHA-256 resume deduplication

Multimodal Learning Disability Detection — **2024**

- 45 classification models across 9 algorithms on 7,000+ record 2019 NHIS dataset (1:8 class imbalance); 86.5% balanced accuracy, 86.1% clinical sensitivity via Random Forest + SMOTE; engineered polynomial interaction terms, missing indicators, and domain composites optimized for clinical significance

EXPERIENCE

Boston University Housing — **Jan 2025 - Present**
Mailroom Assistant — *Boston, MA*

- Built Python automation tool (ReportLab, PostgreSQL) eliminating manual mail processing, reducing turnaround by 26%; in active production use 15+ months with zero downtime

Jawaharlal Nehru Technological University — **Aug 2023 - Jun 2024**
Research Intern - ML Engineer — *Hyderabad, India*

- Trained and deployed YOLOv5 vehicle detection model (PyTorch, OpenCV) on edge hardware for real-time automated traffic signal control; campus-wide rainwater management system (3 Raspberry Pi units, 70-acre campus, 4-week live trial) reduced flood response time by 30%

International Institute of Information Technology Hyderabad — **May 2023 - Aug 2023**
Research Intern - Speech Processing Lab — *Hyderabad, India*

- Designed novel prosody dataset applying audio source separation (Spleeter) informed by music production expertise - cross-domain research contribution reducing preprocessing time by 25%, improving speech translation accuracy 8–10%; presented to cohort of 20+ researchers

EDUCATION

Boston University — **Aug 2024 - May 2026**
Master of Science, Computer Science - ML Engineering & Data Analytics — *Boston, MA*

Jawaharlal Nehru Technological University Hyderabad — **Aug 2020 - Jul 2024**
Bachelor of Technology, Computer Science - AI & Machine Learning — *Hyderabad, India*

TECHNICAL SKILLS

- **Languages**: Python, JavaScript, TypeScript, SQL, R, Java, C/C++
- **NLP & GenAI**: LangChain, Hugging Face Transformers, OpenAI API, Retrieval-Augmented Generation (RAG), BERT Fine-tuning, Named Entity Recognition (NER), Large Language Models (LLMs), spaCy, FAISS, Vector Search, Transfer Learning
- **ML / DL**: PyTorch, TensorFlow, scikit-learn, XGBoost, Random Forest, Interpretable ML, MLflow, Optuna, Neural Networks, Feature Engineering, SMOTE
- **MLOps & Cloud**: AWS Bedrock, GCP, Docker, FastAPI, REST APIs, CI/CD, Jenkins, Redis, PostgreSQL, Supabase, Git, Linux, WebSockets
- **Web & Full Stack**: React, Node.js, Express, Vite, Tailwind CSS, HTML/CSS
- **Computer Vision**: OpenCV, YOLOv5, Tesseract OCR